Saturated Pell certificates and quotient recovery

Eight explicit Diophantine equations for syntactic consistency of ZFC

Research contribution prepared for Vladimir Reshetnikov's ProveIt project
AI-assisted mathematical construction and exact independent audits

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Abstract

We give explicit, materialized integer polynomials with new variable–degree–cost tradeoffs for the absence of a finite ZFC refutation. The main nine-positive construction has exact total degree 999 518; a shorter companion uses 435 binary arithmetic operations at degree 1 628 768. A strong-Pell growth argument gives eleven positive unknowns at degree 213 in 112 operations. For unrestricted integer unknowns, quotient recovery and a separated rational norm give seventeen unknowns at degree 88 and sixteen at degree 184; a degree-ten companion uses twenty-three integers.

The improvements are complete ordinary arithmetic reductions. They compose with explicitly identified inherited source/compiler and logical-calibration theorems; they do not assume consistency. Every fixed integer is supplied, and the archive includes literal division-free circuits, exact algebra certificates, ordinary proofs, and standard-library verifiers. The positive majorant is certified by an identity with 36,309 positive coefficients. We do not claim a global minimum, external historical priority, or a new end-to-end proof-assistant derivation inside the encoded ZFC calculus.

1 Outcome, target, and comparison

1.1 The precise logical target

Fix the repository's syntactic ZFC axiom predicate and proof calculus. Let $\mathrm{Ref}_{\mathsf{ZFC}}(p)$ mean that p codes a finite derivation of falsity, with all axiom and inference checks included. We use

$$\mathrm{Con}(\mathsf{ZFC}) \;\equiv\; \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathsf{ZFC}}(p).$$

The desired conclusion for a fixed polynomial P and an explicitly declared witness domain $\mathcal{D}$ is

$$\mathsf{ZFC} \;\vdash\; [(\neg\exists \mathbf{x} \in \mathcal{D}\ P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC})]\,. \tag{1.1}$$

This is a proof of an equivalence, not a proof of either side. In particular, there is no appeal to the consistency or soundness of ZFC as a premise. The relevant construction proves an existential equivalence with a finite refutation and then negates it.

There is no single established numerical meaning of *simplest*. A smaller number of unknowns may require a higher degree; a lower degree may need a larger arithmetic circuit; compact native constants can require a different source packing. We therefore provide the measured alternatives in Table 1, instead of asserting a nonexistent optimality theorem. The distinction between positive and unrestricted integer witnesses is substantive and is retained in every claim.

For a readily inspectable equation over unrestricted integers, P_{17} in Section 5 is a useful primary presentation: five squared residuals, seventeen unknowns, exact degree 88. If one fewer integer unknown matters more than degree, Section 6 gives the sixteen-variable alternatives. If the number of positive unknowns is the leading measure, Section 3 supplies the nine-variable constructions. These choices do not displace shorter higher-variable equations already present in the repository.

Construction	Witness domain	Unknowns	Exact degree	Operations
Complete-radical H1868	$\mathbb{Z}_{>0}$	9	**999 518**	1086
Simple H1868	$\mathbb{Z}_{>0}$	9	1 474 688	449
Simple compact H2082	$\mathbb{Z}_{>0}$	9	1 628 768	**435**
Strong-Pell weight	$\mathbb{Z}_{>0}$	11	**213**	112
Rational norm, product finish	$\mathbb{Z}$	16	**184**	135
Rational norm, squared finish	$\mathbb{Z}$	16	192	134
Quotient recovery	$\mathbb{Z}$	17	**88**	114
Low-degree synthesis	$\mathbb{Z}$	23	**10**	131

Table 1: Complete new polynomials. All degrees count only the listed unknowns, with the supplied coefficient tuple fixed.

1.2 Measured improvements over the inspected inputs

The reviewed repository commit is

`c1a102936aca437e9999530a0111e6ddab56dc6d`.

The review includes all five archives then present in `docs/incoming`; their exact byte hashes are recorded in `provenance/SOURCES.json`. The comparisons below refer to these fixed inputs, not to a continually changing branch.

Comparison	Inspected input	New construction
Nine positives, small circuit	Degree 1 845 940, 440 operations [1]	Degree 1 628 768, 435 operations
Nine positives, lower degree	Degree 1 158 228, 1753 operations [5]	Degree 999 518, 1086 operations
Eleven positives, centered	Degree 257, 114 operations [3]	Degree 213, 112 operations
Integers, fixed degree 88	19 unknowns, 109 operations [3]	17 unknowns, 114 operations
Integers, fixed degree ten	24 unknowns [4]	23 unknowns, 131 operations

Table 2: Specific comparisons. A variable reduction can increase the operation count; it is not described as improvement in every measure.

The incoming standalone report also gives a nine-positive degree 1 158 240 equation in 811 operations [6]. The new degree-999 518 form lowers its reported degree but uses more operations. That standalone report contains no executable companion in its archive, so its operation figure is treated as a reported comparator. The incoming eleven-positive degree-277 / 111-operation equation remains one operation shorter than our degree-213 form [3]. The active size plan also retains a fifteen-positive degree-157 equation in only 66 operations [1]; we make no claim to beat that total operation count.

1.3 Counting rules and fixed coefficients

The operation measure counts each live binary addition, subtraction, and multiplication in the delivered straight-line circuit once. A square is one multiplication. Multiplication by a fixed scalar is also counted. Intermediate registers can be reused. Fixed integer leaves are free in this arithmetic-circuit measure, so their storage cost is stated separately. No division, variable exponent, inequality test, logarithm, bit operation, or proof predicate occurs in a delivered polynomial evaluator.

The following are the native coefficient costs, not the coefficients of a fully expanded enormous polynomial:

Coherent tuple	Ports	Raw bytes	Largest port, bits
Compact54-sharp (nine, H2082)	4	215492	434054
Original31 (nine, H1868)	4	533099	1070004
Compact44 (eleven positives)	4	295388	592884
Source34 plus determinant scales	6	615883	1065043

The full raw and compressed costs, individual bit lengths, and raw hashes are in `COEFFICIENT_COSTS.json`. The nine-variable circuits also materialize derived kernel coefficients and power-of-two scale leaves; their bindings and actual bytes are supplied separately. These are polynomials with specified finite integer constants, not a universal polynomial with an unspecified program-index parameter. The coherent tuples must not be mixed merely because two radix exponents happen to agree.

For witnesses in $\mathbb{N} = \{0, 1, \ldots\}$, any positive-domain construction can be translated by $x_j = u_j + 1$. This preserves total degree and existence equivalence. The reported operation counts are for the original positive circuits; the translated circuit is a different counted artifact. Likewise, the integer constructions are not obtained by silently allowing negative inputs in a positive-domain theorem.

2 Proof architecture and inherited dependencies

2.1 What the ordinary proof composes

Each component establishes, in its stated domain,

$$\exists \mathbf{x}\, P_{\text{new}}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists \mathbf{y}\, \mathcal{S}_{\text{authenticated}}(\mathbf{y}) = 0.$$

The authenticated arithmetic sources have inherited ordinary theorems relating their zeros to the fixed native computation and then to $\exists p\ \mathrm{Ref}_{\mathsf{ZFC}}(p)$. Composition gives (1.1). The archive retains the relevant source proofs, the concrete coefficient adapters, and their provenance. It does not re-prove the entire earlier machine/tableau/calculus chain.

In the nine-positive branch the inherited interfaces are the native positive source, the exact sparse carry packing, the numerical ratio theorem, and the old ProductPell certificate. We prove the new saturated certificate and both complete finalizers. The eleven-positive branch inherits a weighted native Pell source and its canonical weight lifting; we prove the smaller weight works before the required source positivity has been established. The integer branch inherits signed orientation, strong-rank recovery, and canonical determinant completeness; we prove the quotient elimination and subsequent norm merger without assuming either omitted equation in their own justifications.

These are ordinary unbounded proofs. The hypothetical proof or computation can be arbitrarily long. The arguments use finite polynomial identities, integer divisibility, induction on recurrence indices, and standard Pell and three-square results available in ZFC. They never infer a universal result from a bounded search.

2.2 Three different verification artifacts

It is necessary to distinguish:

1. An ordinary mathematical proof of the new reductions, composed with the named ordinary source/calibration lemmas.
2. Exact finite verification of the delivered identities, constants, circuits, source restorations, and degree certificates.

3. A proof-assistant theorem for the entire new equation, or an actual derivation inside the repository's encoded object-language ZFC.

This contribution supplies the first two. It does not supply the third. The exact K215 machine already has external Lean and Rocq correctness theorems [1]; the recent native arithmetic/compiler and internal-ZFC formalization have distinct remaining obligations. In particular, an external theorem about the standard arithmetic predicate is not automatically an exported object-language proof covering every internal natural of a nonstandard model.

The retained `provenance/OBJECT_ZFC_AT_PIN.md` states that boundary precisely. We do not promote the finite Python audits, or a general claim of formalizability, into a newly checked internal biconditional. The ordinary theorem in this report is a composition with identified inherited mathematical lemmas; it is not a fresh end-to-end kernel certificate. The repository's current size-reduction plan explicitly pauses extension of Lean formalization [1].

2.3 Scope of the new ideas

The incoming round-11 work already used norm saturation for a different ten-integer Sun/Lucas construction [2]. Incoming papers also used whole-radical approximation, root centering, determinant guards, joint positivity, and adjacent-coefficient rational Pell descent [5, 4, 3]. Those ideas are credited.

The new content is the positive ProductPell saturation theorem, its exact full-radical correction and measured complete equations; the strong-Pell growth bound replacing a larger guard by $i^3 + cT$; the explicit removal of a quotient witness after a noncircular bootstrap; and the separated weighted rational norm that removes one further integer coordinate. The lower-degree twenty-three-integer equation is a new composition of already available local transformations. These are precise improvements relative to the reviewed sources. No claim of historical priority beyond that comparison is made.

3 Positive Pell saturation and a degree below one million

This section gives three explicit polynomials in the same nine strictly positive unknowns

$$b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}.$$

Their other registers are polynomial abbreviations, not additional existential variables. The finite source tables, fixed integer payloads, and division-free arithmetic circuits accompany the report.

Polynomial	Exact degree	Binary operations
Compact H2082, simple majorant	1 628 768	435
H1868, simple majorant	1 474 688	449
H1868, complete-radical majorant	**999 518**	1086

An operation is one addition, subtraction, or multiplication; squaring and scalar multiplication each cost one, shared registers are reused, and fixed integer leaves cost zero. These are different points in a degree–cost comparison. The first improves the inspected 440-operation H2082 equation; the last improves the inspected nine-positive degree frontier, including the incoming radical-guard construction [1, 5, 6]. The variable count is unchanged.

The incoming round-11 package already applies *norm saturation* to a different ten-integer Sun/Lucas certificate [2]. We credit that idea explicitly. The new steps here are its transfer to the positive ProductPell certificate, a majorant of the complete radical DFI, and their composition with the H1868 source. The whole-radical approximation viewpoint also builds on the incoming majorant work [5, 6]. No historical priority or global minimum is claimed.

3.1 Removing one square factor from the positive certificate

For $A \geq 2$, let

$$\psi_A(0) = 0, \qquad \psi_A(1) = 1, \qquad \psi_A(t+2) = 2A\psi_A(t+1) - \psi_A(t),$$

and let χ_A have initial values $1, A$ and the same recurrence. Thus

$$\chi_A(t)^2 - (A^2-1)\psi_A(t)^2 = 1.$$

We use the standard positive Pell parametrization and the divisibility and congruence lemmas used in Jones's ProductPell proof [11, 10].

Lemma 3.1 (Saturated positive ProductPell)**.** *Let $A \geq 2$, $B \geq 3$ be odd, $C \geq B$, and $J \geq 1$. For $i, j \in \mathbb{Z}_{>0}$, define*

$$\begin{aligned} D &= (A^2-1)C^2 + 1, & r &= iJC^2, \\ F &= (A^2-1)Dr^2 + 1, & L_m &= CD + 1 - A, \\ G &= A + FL_m, & H &= C + F\big(B + (j-1)C\big), \\ I &= (G^2-1)H^2 + 1. \end{aligned} \tag{3.1}$$

Then

$$C = \psi_A(B) \quad \Longleftrightarrow \quad \exists i, j \in \mathbb{Z}_{>0} \;\; DFI \text{ is an integer square.}$$

Proof. All displayed radicands are positive. The defining equations give

$$\begin{array}{lll} F \equiv 1 \pmod{D}, & G \equiv 1 \pmod{D}, & I \equiv 1 \pmod{D}, \\ F \equiv 1 \pmod{C}, & G \equiv 1 \pmod{C}, & H \equiv B \pmod{C}, \\ G \equiv A \pmod{F}, & H \equiv C \pmod{F}, & I \equiv D \pmod{F}. \end{array}$$

Consequently D, F, I are pairwise coprime. A square product therefore makes each factor a square. Write $D = d^2$, $d > 0$, and introduce only inside the proof

$$E' = iJdC^2.$$

Then $F = (A^2-1)E'^2 + 1$ and $C^2 \mid E'$. Pell parametrization supplies

$$C = \psi_A(t), \qquad E' = \psi_A(u), \qquad H = \psi_G(v).$$

Here $2 \leq t < C$, and the square-divisibility lemma

$$\psi_A(t)^2 \mid \psi_A(u) \implies \psi_A(t) \mid u$$

gives $C \mid u$. Since $F = \chi_A(u)^2$, recurrence congruence and $H \equiv C \pmod{F}$ imply

$$\psi_A(v) \equiv \psi_A(t) \pmod{\chi_A(u)}.$$

The Pell step-down congruence gives $v \equiv \pm t \pmod{2u}$, and hence $v \equiv \pm t \pmod{C}$. But $G \equiv 1 \pmod{C}$ gives $\psi_G(v) \equiv v \pmod{C}$, so $B \equiv \pm t \pmod{C}$. The bounds leave $B = t$ or $B + t = C$. The latter is impossible: $\psi_A(t) \equiv t \pmod{2}$ makes $C - t$ even, whereas B is odd.

For completeness suppose $C = \psi_A(B)$, and put $d = \chi_A(B)$. Periodicity of the invertible Pell recurrence modulo JDC^2 supplies a positive index u and $i_{\text{old}} > 0$ with

$$\psi_A(u) = i_{\text{old}} JDC^2.$$

Set $i = d\, i_{\mathrm{old}}$. Then the new F equals

$$(A^2-1)(i_{\mathrm{old}} JDC^2)^2 + 1 = \chi_A(u)^2.$$

Put $H_0 = \psi_G(B)$. Recurrence congruences give $H_0 \equiv C \pmod F$ and $H_0 \equiv B \pmod C$. Because $F \equiv 1 \pmod C$, the integer $q = (H_0 - C)/F$ satisfies $q \equiv B \pmod C$. Moreover $L_m \geq C$, $G \geq FC$, and $\psi_G(B) \geq BG$ for $B \geq 2$; thus

$$H_0 \geq BFC \geq FB + C, \qquad q \geq B.$$

Therefore $j = 1 + (q-B)/C$ is positive and realizes the displayed H. Finally $I = \chi_G(B)^2$, proving the square-product condition. □

The earlier positive construction used

$$F_{\mathrm{old}} = (A^2-1)(i_{\mathrm{old}} JDC^2)^2 + 1.$$

Its proof used $D \mid E_{\mathrm{old}}$ only to obtain $F \equiv 1 \pmod D$. The new polynomial provides that congruence directly, while retaining $C^2 \mid E'$, the divisibility actually needed by the index argument. The completeness map $i = d\, i_{\mathrm{old}}$ is an embedding, not a claimed bijection of all witness tuples.

3.2 The H1868 source as a positive-domain prefix

The source data are pinned to repository snapshot `c1a102936aca` (the full hash is in the provenance files) and the incoming round-11 H1868 certificate [1, 2]. They retain the complete original31 coefficient tuple. Its four fixed integers are materialized in the archive; they are neither free parameters nor unspecified program indices. The compact H2082 companion instead retains its own coherent compact tuple from the source-54 lineage [7].

For H1868 the 31 coordinate weights, in the supplied source order, are

$$(1, 7, 61, 36, 16, 85, 20, 67, 43, 11, 34, 13, 96, 80, 29, 50,\\ 2, 109, 110, 112, 116, 91, 53, 55, 27, 101, 92, 9, 107, 37, 60).$$

The 22 target positions, in source-row order, are

$$(1141, 28, 1437, 1069, 1836, 1354, 1814, 614, 267, 1408, 1403,\\ 889, 337, 1587, 413, 673, 1139, 772, 1131, 1018, 1055, 85).$$

The literal files `nine/data/source.json`, `nine/data/code.json`, and `nine/data/kernels.json` specify the source residuals ρ_j and the three integer polynomials D_0, D_1, D_2. They satisfy, as exact polynomial identities,

$$[T^{q_j}]\left(D_2(T)U_{\mathbf{p}}(T)^2 + D_1(T)U_{\mathbf{p}}(T) + D_0(T)\right) = 2\rho_j(\mathbf{p}), \qquad U_{\mathbf{p}}(T) = \sum_{\ell} p_\ell T^{w_\ell}. \tag{3.2}$$

Equal kernel coefficients at a shared position are retained once. The audit checks all absent monomials as well as the intended ones: 22 identities, 20,466 convolution interactions, and 81 stored positions. The kernel degrees are $(1408, 1727, 1636)$, with leading coefficients $(-6, 2, 1)$; therefore the convolution height is 1868, the maximum weight is 116, and the maximum target is 1836.

Put $\kappa = 2134939$. Define

$$\begin{aligned}
B_c &= 2^{\kappa} b^2, & \ell &= \sum_{\ell} B_c^{w_\ell}, & T_c &= \sum_{j} B_c^{q_j},\\
U_s &= \ell + g, & G_s &= (D_2(B_c)U_s + D_1(B_c))U_s + D_0(B_c),\\
N_0 &= B_c^{117}, & M_c &= N_0 - 1 - (b-1)\ell,\\
S_c &= g + N_0(2G_s + T_c), & T_m &= M_c + N_0(B_c - 2)T_c,\\
L_c &= 2bB_c^{1986}, & R &= L_c(S_c + T_m + 1) - S_c - 1,\\
N &= 2^{16012032} B_c^{1055}, & t_c &= bB_c^{116}.
\end{aligned} \tag{3.3}$$

The factor 2 in L_c, and the absence of an additional b in N, are essential.

Here every kernel coefficient has magnitude below $2^{2134927}$. The quadratic layer dominates the linear and constant layers by 25 and 460 powers of B_c, respectively. The inherited geometric-series and sparse-carry bounds consequently give $G_s > 0$, $R, N \geq 8$, and $b \leq R, N$ on all positive inputs, before the base is known to be binary. On the recovered range $g < 4t_c$, they give

$$S_c + T_m < B_c^{1986} < L_c.$$

The coefficient bound and the unchanged 32-position target headroom are exactly the hypotheses used by the inherited sparse-carry proof [2, 9].

For clarity, the binary threshold can be recalculated directly. If $b = 2^k$, $B_c = 2^v$, and $L_c = 2^u$, then

$$v = \kappa + 2k, \quad u = 1 + k + 1986v, \quad \operatorname{pop}(T_m) = 139v - 31k - 22.$$

The two-block identity for R gives

$$\operatorname{pop}(R) = u + \operatorname{pop}(T_m) - \tau, \qquad \tau = \operatorname{pop}(S_c) + \operatorname{pop}(T_m) - \operatorname{pop}(S_c + T_m) \geq 0.$$

Moreover

$$u + \operatorname{pop}(T_m) = 2125\kappa + 4210k - 21 = 2\log_2 N.$$

Legendre's identity therefore gives

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \text{ AND } T_m = 0. \tag{3.4}$$

The sparse-carry theorem decodes the right side into a positive common source zero. Conversely a fixed source zero packs for all sufficiently large binary b, with $0 < g < t_c$. Positivity of g follows because the all-one source tuple is not a zero: its fifth residual equals 1. The endpoint $b = 1$ cannot create a spurious solution, since its low mask would force $g = 0$.

3.3 The common ratio registers

From either authenticated prefix define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M &= RY,\\
P &= 2M^2 U, & K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= bw, & J &= 4A - 5,\\
Q &= 3W_1 C - 2(W_1^2 - 1), & a_0 &= (P^2 - 1)K^2 + 1,\\
m &= K^2(4t_c - g) - 16t_c(C - KY)^2, & \Omega &= Jm,\\
E_0 &= J(n + W_1^2) + Q, & V_0 &= PK.
\end{aligned} \tag{3.5}$$

Use (3.1) with $B = B_1$, and abbreviate

$$Z = B_1 + (j-1)C, \qquad a_{1,\mathrm{raw}} = DFI.$$

The inherited numerical ratio theorem converts the source conditions into

$$a_0 \text{ square}, \qquad C = \psi_A(B_1), \qquad J \mid Q, \qquad m > 0,$$

with the weak soundness bound $g < 4t_c$ and the stronger completeness bound $g < t_c$ [11, 9, 10]. When that theorem is phrased using the old ProductPell witnesses, the equivalence is interpreted existentially. After recovering $C = \psi_A(B_1)$ we may reconstruct the old witnesses. We do not divide an arbitrary new i by $\sqrt{D}$.

3.4 Two exact majorants

The simple choice is

$$T_{\mathrm{simple}} = ADr, \qquad V_1 = T_{\mathrm{simple}}GH, \qquad a_1 = a_{1,\mathrm{raw}}. \tag{3.6}$$

Its unconditional certificate is particularly short:

$$\begin{aligned} T_{\mathrm{simple}}^2 - DF &= D(Dr^2 - 1) > 0, \\ V_1^2 - a_{1,\mathrm{raw}} &= D(Dr^2-1)G^2H^2 + DF(H^2-1) > 0. \end{aligned} \tag{3.7}$$

This supplies the 435-operation compact-H2082 polynomial and the 449-operation H1868 polynomial in the table.

For the lower-degree construction we approximate the *complete* radical $\sqrt{DFI}$. Here is an exact finite definition of the supplied 333-term polynomial, so its construction does not depend on a numerical approximation. Regard

$$\mathcal{R}(A) = D(A)F(A)I(A) = \sum_{k=0}^{36} q_k(C,i,Z)A^k$$

as a polynomial in A, with $J = 4A - 5$. Its leading coefficient is $2^{20}C^{38}i^{10}Z^2$. In the rational function field $\mathbb{Q}(C,i,Z)$, set

$$t_{18} = 1024C^{19}i^5Z, \qquad t_{18-r} = \frac{q_{36-r} - \sum_{s=1}^{r-1} t_{18-s}t_{18-r+s}}{2t_{18}} \quad (1 \le r \le 18). \tag{3.8}$$

Put

$$\begin{aligned} \sigma &= 1024C^5 i, \\ \mathcal{P}_0 &= \sigma \sum_{k=0}^{18} t_k A^k, \\ \mathcal{P} &= \mathcal{P}_0 + 297874C^{24}i^6Z. \end{aligned} \tag{3.9}$$

The finite coefficient calculation proves $\mathcal{P}_0, \mathcal{P} \in \mathbb{Z}[A,C,i,Z]$; the authoritative literal integer table is `nine/data/saturated_corrected_P.json`. The divisions in (3.8) describe a finite way to derive that fixed table. The delivered evaluator uses only its integer coefficients, addition, subtraction, and multiplication.

Proposition 3.2 (Complete-radical certificate)**.** *For $A \ge 3$, $C, i, Z \ge 1$,*

$$\mathcal{P} > \sigma\sqrt{DFI} > 0.$$

Proof. The independent exact checker reconstructs DFI from its defining polynomials and verifies

$$\left(\mathcal{P}^2 - \sigma^2 DFI\right)(x+3, y+1, u+1, v+1) = \mathcal{C}(x, y, u, v),$$

where the supplied coefficient table for $\mathcal{C}$ has 36,309 strictly positive nonzero coefficients and constant term

$$8341416066461366340.$$

It also verifies that $\mathcal{P}(x+3, y+1, i, Z)$ has positive coefficients and a positive constant term. Thus the defect and the majorant itself are strictly positive on the required domain. These exact polynomial identities prove the inequality. □

For this variant use

$$a_1 = \sigma^2 DFI, \qquad V_1 = \mathcal{P}. \tag{3.10}$$

The finite recurrence, literal coefficient tables, and positive-defect certificate serve different purposes: the recurrence explains the construction; the table defines the exact polynomial; the certificate proves its uniform inequality. The Laurent approximation itself is not used as an asymptotic assumption.

3.5 The single equation and its proof

For either choice (3.6) or (3.10), put

$$\Theta = E_0 - \Omega(V_0 + V_1), \qquad \boxed{\mathcal{F}_9 = \left(\Theta^2 - \Omega^2(a_0 + a_1)\right)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{3.11}$$

Theorem 3.3. *The polynomial $\mathcal{F}_9$ has a positive integer zero if and only if its authenticated positive source has a zero. With the inherited source-calculus calibration,*

$$\mathrm{ZFC} \vdash \left(\neg\exists (b, g, h, s, w, C, i, j, n) \in \mathbb{Z}^9_{>0}\ \mathcal{F}_9 = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Proof. The unconditional source bounds give $A \geq 33280$, $A > W_1$, and $J > 3A > 0$. Also

$$A^2 - P = M^2(U^2 + 1) \geq 1, \qquad D \geq A^2, \qquad D > C^2.$$

Both radicands are positive. We have $V_0^2 - a_0 = K^2 - 1 > 0$; the chosen majorant gives $V_1 > \sqrt{a_1}$. Furthermore

$$E_0 = Jn + (J-2)W_1^2 + 3W_1 C + 2 > 0.$$

If $m = 0$, then $\mathcal{F}_9 = E_0^4 > 0$. Otherwise factoring the two-root norm over the reals gives signs ϵ_0, ϵ_1 with

$$\frac{E_0}{J} = m\left(V_0 + V_1 + \epsilon_0\sqrt{a_0} + \epsilon_1\sqrt{a_1}\right).$$

The parenthesis is positive for every sign pair. Thus $m > 0$ *before* any Pell-index or square-root-integrality argument.

The margin now gives $|C - KY| < K/2$, and therefore

$$C > 2K > 2R + 1 + W_1 > B_1.$$

Since $D > PC^2$, $F > D$, and $C > K$,

$$a_{1,\mathrm{raw}} > D^2 > P^2 K^2 \geq a_0.$$

This also separates the radicands in the scaled variant. Hence Θ/Ω, a signed sum of their roots, is rational and nonzero. For signed roots u_0, u_1,

$$2(u_0+u_1)u_0 = (u_0+u_1)^2 + a_0 - a_1$$

makes both roots rational, then integral. In the complete variant, $\sqrt{a_1}/\sigma$ is a rational square root of the integer DFI, so it too is integral. The same norm-factor identity yields $J \mid E_0$, and hence $J \mid Q$. Lemma 3.1 gives $C = \psi_A(B_1)$. The ratio theorem and the authenticated carry test now recover a source zero. If necessary, reconstruct the old ProductPell witnesses before invoking the original ratio theorem; the other ratio registers remain fixed.

Conversely a source zero supplies the ratio conditions, with $m > 0$. Lemma 3.1 supplies positive i, j making DFI a square. Let z_0, z_1 be the positive integral roots of the chosen a_0, a_1, and define

$$n = m(V_0 + V_1 + z_0 + z_1) - W_1^2 - Q/J.$$

This is an integer. It is positive because

$$W_1^2 + Q/J = (1 - 2/J)W_1^2 + (3W_1C + 2)/J < 2D,$$

whereas $V_1 > 4D$. For the simple majorant the latter follows directly from $V_1 = ADrGH$. For the complete majorant, $F > 3D$, $G \geq 3$, and $H \geq 2$ give $DFI > 99D^2$, and Proposition 3.2 applies. This choice of n makes one norm factor vanish. The existence equivalence follows, and negating it gives the displayed consistency equivalence. □

3.6 Why the degree is exactly 999,518

The complete-radical construction needs a cancellation-aware calculation. For H1868 the positive leading terms of the source are

$$R \sim 4bB_c^{3971}, \qquad N = 2^{16012032}B_c^{1055}.$$

Thus, writing $r_R = \deg R$, $\nu = \deg N$, and $a = \deg A$,

$$r_R = 7943, \qquad \nu = 2110, \qquad a = r_R + 4\nu + 2 = 16385.$$

For weights $(a, 1, 1, r_R)$ on (A, C, i, Z), the unique leading monomials of the complete majorant and its defect are

$$\begin{aligned} \mathcal{P} &: \quad 1048576A^{18}C^{24}i^6Z, \\ \mathcal{P}^2 - \sigma^2 DFI &: \quad 624687054848A^{18}C^{48}i^{12}Z^2. \end{aligned}$$

Consequently

$$\deg V_1 = 18a + r_R + 30 = 302903, \qquad \deg \delta_1 = 18a + 2r_R + 60 = 310876,$$

where $\delta_1 = V_1^2 - a_1$. Also, for $\delta_0 = V_0^2 - a_0$,

$$\begin{aligned} \deg \Omega = 5a + 2W + 3, \quad \deg E_0 = a + 4, \\ \deg V_0 = 3a + r_R + 1, \quad \deg \delta_0 = 3a + r_R + 2. \end{aligned}$$

Put

$$\mathcal{B} = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1).$$

Its unique leading contribution is $\Omega^2\delta_1$. The exact identity

$$\mathcal{F}_9 = \mathcal{B}^2 + 4\Omega^2 V_0 V_1 \mathcal{B} + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1)$$

has four displayed contribution degrees

$$950392, \qquad \boxed{999518}, \qquad 991546, \qquad 753714.$$

The maximum is unique and its leading coefficient is positive. Therefore

$$\deg \mathcal{F}_9 = 59a + 4r_R + 8W + 103 = 999518$$

is attained, not merely an upper bound from a circuit.

Verification and proof boundary. The independent standard-library audit reconstructs the source identities, the positive population threshold, the 3035-term local radicand, the 333-term majorant, the complete positive-defect certificate, and the exact degree comparison without importing the producer. A separate literal circuit checker verifies the exported equations and operation counts. The three complete Pell fixtures and 5472 local arithmetic checks support transcription checks; they are not experiments concerning ZFC's consistency. The consistency theorem uses the identified ordinary source/compiler calibration [1, 9]. This work supplies no new end-to-end Lean/Rocq certificate or exported derivation in an internal ZFC calculus.

4 An eleven-positive equation of degree 213

This companion lowers the incoming centered eleven-positive equation from degree 257 and 114 operations to degree 213 and 112 operations. The incoming degree-277 equation still has one fewer operation [3]. The new mechanism is to use growth forced by the strong Pell equation to bound the transport guard, then center the norm at *both* the strong and auxiliary roots. All statements in this section concern ordinary proofs formalizable in ZFC. The compact44 native compiler and the normalized source calibration are inherited; this is not a new end-to-end proof-assistant certificate.

4.1 The complete polynomial

The unknowns are

$$(j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma) \in \mathbb{Z}_{>0}^{11}.$$

Use the entire compact44 coefficient tuple (β, K_0, M_C, M_F), with $\beta = 2^{590069} - 1$ and the already shifted M_F. The companion includes their actual integer payloads and binding; these ports are not interchangeable with a different native tuple. Every following name abbreviates a polynomial and adds no unknown:

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H, & \ell &= EkY, & I &= ic^2, & D_f &= 1 + \Delta I^2,\\
Q &= \Delta^2 I^2, & \delta &= Q - 1, & M &= q^3(q-1).
\end{aligned}$$

Further put

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
B_t &= (qK_0 + X)C - qF, & D_\tau &= \ell(\ell + k) + 1,\\
V &= D_f(cT - R) - c, & \bar{f} &= (a+2)I,\\
h_* &= i^3 + cT, & W &= 1 + (q-1)h_*,\\
\bar{Z} &= M(\ell + k + n + cT + V + \bar{f}) + (q-1)Y - q^3 W B_t.
\end{aligned}$$

The unchanged main carrier is

$$\begin{aligned}
v &= k - R - 1, & S &= 4qY, & U &= qS,\\
D_* &= U(X + ac) + H(U\gamma + SX - v),\\
A_1 &= D_*^2 - U^2(1 + \Delta c^2).
\end{aligned}$$

The octic norm is evaluated by

$$\begin{aligned} A_r &= M^2 D_\tau, & B_r &= M^2 D_f, & C_r &= M^2(QV^2-1),\\ T_N &= \delta^2 A_r B_r, & R_N &= \delta C_r, & & \\ J_N &= \delta(\bar{Z}^2 - A_r - B_r) - C_r, & & & & \\ N_c &= J_N^2 - 4\{T_N + R_N(A_r + B_r)\}, & & & & \\ N_8 &= N_c^2 - 64 T_N R_N \bar{Z}^2. & & & & \end{aligned}$$

The single equation is

$$\boxed{P_{11} = N_8 + A_1 + N_8 A_1 = 0.} \tag{4.1}$$

It uses only fixed integer coefficients, fixed exponents, addition, subtraction, and multiplication. Since $4 \mid U$, both A_1 and N_8 are squares modulo 4. Thus $(1+N_8)(1+A_1) = 1$ forces both factors to be 1: the alternative $(-1,-1)$ would require residues -2. Consequently (4.1) is exactly the conjunction $N_8 = A_1 = 0$ on its stated domain.

4.2 Source boundary and recovery before positivity

The inherited weighted source theorem [8, 3] concerns positive coordinates satisfying the main residual, the strong norm, the auxiliary norm, and the first carrier

$$\begin{aligned} f^2 - \Delta(ic^2)^2 &= 1,\\ QV^2 - (Q-1)(V+u)^2 &= 1,\\ Z^2 &= M^2 D_\tau,\\ Z &= M(\ell + k + r) + (q-1)Y - q^3 W B_t. \end{aligned}$$

Its soundness holds for every positive integer h_*. For canonical witnesses, after the strong and auxiliary pairs have been chosen, a chosen weight can be lifted by

$$r = W t_0 + \tau - \ell - k - s > qW - 2k, \qquad t_0 \geq q, \quad 0 < s < k, \quad \ell < \tau < \ell + k.$$

The source has a positive zero precisely when the fixed ZFC calculus has a finite refutation. The authentic mask bounds are

$$K_0 \geq 1, \quad 0 < M_C < \beta, \quad 0 < M_F < 2\beta, \qquad 0 < (M_C + qM_F)J < 2q^2.$$

Once $C > 0$, these imply $3q < R < q^4 - 2$.

Here is the part of the norm recovery that must precede $C > 0$. For every positive input, $I \geq c^2$, $\Delta \geq 15$, and

$$1 < D_\tau < D_f < Q, \qquad |V| > 1.$$

The rational radicand $D_y = (QV^2-1)/(Q-1) > V^2$ is distinct from D_τ, D_f. Moreover, with $t = \sqrt{D_\tau}$, $f = \sqrt{D_f}$, $y = \sqrt{D_y}$, no signed sum of t, f, y is zero. If $cT - R = 0$, then $f > t + y$; otherwise $y > f + t$. These inequalities depend only on $I \geq c^2$, not on C or a bound on R; their elementary proof is retained in the companion source audit.

The universal three-root norm therefore gives, at $N_8 = 0$,

$$\bar{Z}/M = \tau_* + f_* + y_*, \qquad \tau_*^2 = D_\tau, \quad f_*^2 = D_f, \quad y_*^2 = D_y. \tag{4.2}$$

All three roots are rational. One can prove this without assuming independent square classes: for signed roots x_1, x_2, x_3 with rational nonzero sum t_0, put

$$u_0 = (t_0^2 - x_1^2 - x_2^2 - x_3^2)/2, \qquad v_0 = \frac{u_0^2 - x_1^2 x_2^2 - x_1^2 x_3^2 - x_2^2 x_3^2}{2t_0}.$$

Then u_0 and v_0 are rational and $x_1 = x_1^2(x_1^2+u_0)/(x_1^2t_0+v_0)$. The denominator is $x_1(x_1+x_2)(x_1+x_3) \neq 0$, by distinctness of the radicands; use the cyclic formulas as well. Thus τ_*, f_* are integers. The adjacent-coefficient Pell descent from $QV^2-(Q-1)y_*^2=1$ makes y_* an integer too [3]. In particular $M \mid \bar{Z}$.

Reduction modulo q^3 and $q-1$ now gives $q^3 \mid Y$ and $q-1 \mid B_t$, independently of the sign of C. The main residual makes $g = |D_*/U|$ an integer and $g^2 - \Delta c^2 = 1$. Writing

$$\chi_A(r) + \psi_A(r)\sqrt{A^2-1} = (A+\sqrt{A^2-1})^r,$$

Pell classification gives $c = \psi_{a+2}(p)$ and $k = 2\psi_{2XY^2+1}(m_1)$. Here $p \geq 2$ and $m_1 \geq 2$: $m_1 = 1$ would give $k = 2$, $c = 2Y+1 < 2(a+2)$. Consequently, still before proving $C > 0$,

$$c > kY \geq 4(2XY^2+1)Y > 8XY^3 \geq 8q^9. \tag{4.3}$$

4.3 The new conditional growth bound

Lemma 4.1 (Strong-Pell growth budget)**.** *Suppose $q \geq 2$, $Y \geq q^3$, $A \geq 2Y+2$, $\Delta = A^2-1$, $c = \psi_A(p) \geq 6$, and positive integers f, i satisfy $f^2 - \Delta(ic^2)^2 = 1$. Then*

$$i > 8q\Delta c^4, \qquad i^3 > 6qf^2, \qquad i^3 > q^2, \qquad Aic^2 < i^2.$$

If $0 < k < c$, also $i^3 > Aic^2 + k$.

Proof. Classify $f = \chi_A(m)$ and $ic^2 = \psi_A(m)$. Strong divisibility gives $p \mid m$: the addition law and $\gcd(\chi_A(r), \psi_A(r)) = 1$ give the Euclidean identity $\gcd(\psi_A(r), \psi_A(s)) = \psi_A(\gcd(r,s))$. Write $m = pv$ and $g = \chi_A(p)$. Composition gives $ic = \psi_g(v)$. As $g^2 \equiv 1 \pmod{c}$, recurrence gives $\psi_g(v) \equiv vg^{v-1} \pmod{c}$; hence $c \mid v$. Thus $v \geq c \geq 6$ and $ic > g^5$. Since $g^2 = 1+\Delta c^2$ and $\Delta > 4Y^2$,

$$i > \Delta^{5/2}c^4 > 8Y^3\Delta c^4 \geq 8q^9\Delta c^4 \geq 8q\Delta c^4.$$

Putting $H_0 = \Delta c^4 \geq 15$ gives $i^3 > 8qH_0i^2 > 6q(H_0i^2+1) = 6qf^2$. Also $i > q$, $i > Ac^2$, and $i > c > k$. Therefore $Aic^2 < i^2$ and $i^3 - i^2 > k$. □

Apply the lemma after (4.2), with $A = a+2$ and $f^2 = D_f$. It gives

$$i^3 > 6qD_f, \qquad i^3 > q^2, \qquad i^3 > \bar{f} + k. \tag{4.4}$$

Also $\bar{f} > f$, since $\bar{f}^2 - D_f = I^2 - 1 > 0$. These are consequences of the recovered Pell equations; no assertion of these bounds on arbitrary positive inputs is needed.

4.4 The noncircular positivity bootstrap

Suppose $C \leq 0$, put $d = -C$ and $B = -B_t > 0$. Then

$$B = (qK_0 + X)d + qF, \qquad R = [q(q-F)+d](q^2-1) + (M_C + qM_F)J.$$

The mask bounds imply

$$R < qB + q^4 + q^2, \qquad R < 0 \Longrightarrow -R < q^2B. \tag{4.5}$$

Let $s = Y/q^3 > 0$. Equation (4.2) becomes

$$\tau_* + f_* + y_* = \ell + k + n + cT + V + \bar{f} + s + \frac{WB}{q-1}.$$

Its left side is at most $t + f + y$, with $t < \ell + k$. We derive a contradiction in both possible signs of V.

As $\delta > D_f$,

$$0 < y - |V| = \frac{V^2 - 1}{\delta(y + |V|)} < \frac{|V|}{\delta}.$$

If $V > 0$, the integer $b = cT - R$ is positive, so $y - V < b$ and

$$cT + V + \bar{f} - f - y > cT - b = R.$$

For $R \geq 0$ the right side of the radical equality already exceeds $t + f + y$. For $R < 0$, the remaining term exceeds $h_* B \geq i^3 B > q^2 B > -R$, with the same contradiction.

If $V < 0$, then $R \geq cT \geq c > 8q^9$. Thus $q^4 + q^2 < R/2$ and (4.5) gives $R < 2qB$. We have $|V| < D_f R$ and $y - |V| < R$, whence

$$cT + V + \bar{f} - f - y > -(2D_f + 1)R > -3D_f R.$$

But $WB/(q-1) > h_* B \geq i^3 B > 6qD_f B > 3D_f R$, again a contradiction. The case $V = 0$ was excluded earlier. Therefore $C > 0$.

Now $3q < R < q^4 - 2$, and (4.3) gives $c > 2R$. Hence $V > 1$ and the positive-gap estimate strengthens to

$$cT + V + \bar{f} - f - y > R > 0.$$

Restore positive source witnesses by

$$f_{\text{old}} = f, \qquad u = y - V, \qquad r = n + cT + V + \bar{f} - f_* - y_*.$$

Here $u > 0$ and $r > n > 0$ for every permitted choice of signs. Subtracting $M(f_* + y_*)$ from $\bar{Z}$ gives exactly the first source carrier $Z = M\tau_*$. All four source equations hold, so source soundness yields a finite ZFC refutation.

Conversely, choose canonical outer, strong, and auxiliary source witnesses first, then set $h_* = i^3 + cT$. By (4.4), $h_* > cT + \bar{f} + k$; the weighted source construction supplies $r > qW - 2k > cT + \bar{f}$. Take

$$n = r - cT - \bar{f} + f + u > 0.$$

Then $\bar{Z} = M(\tau + f + y)$, so $N_8 = 0$, and the unchanged canonical main carrier gives $A_1 = 0$. No new congruence on the strong pair is required. The witness choices have no dependence loop. Composing both implications with the inherited source calibration proves

$$\mathrm{ZFC} \vdash (\nexists \boldsymbol{x} \in \mathbb{Z}_{>0}^{11}\ P_{11}(\boldsymbol{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}).$$

4.5 Exact degree and reproducible finite evidence

Put $e = I^2 - 1$ and $z = \bar{Z} - M(V + \bar{f})$, so $D_f = \bar{f}^2 - e$ and $\deg z = 9$. Expand the universal norm using

$$x = M(V + t) + z, \quad A = M^2 D, \quad B = M^2(t^2 - e), \quad C_{\text{num}} = M^2(QV^2 - 1), \quad \delta = Q - 1.$$

With weights $(18, 18, 4, 7, 9, 10, 8)$ on (Q, V, M, t, z, e, D), the exact 1110-term integer expansion has degree 200 and leading form

$$64M^6 Q^4 V^4 t^2 z^2. \tag{4.6}$$

The checker expands this identity exactly; naive gate-degree propagation does not suffice. Write $q_0 = 3\beta j$, $k_0 = \eta + \zeta$, and

$$B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

The relevant leading forms are

$$\begin{aligned} D_{f,0} &= i^2k_0^4X^2Y^6, & Q_0 &= i^2k_0^4X^4Y^8,\\ R_0 &= q_0^3(q_0-F), & V_0 &= -D_{f,0}R_0,\\ \bar{f}_0 &= ik_0^2XY^3, & M_0 &= q_0^4,\\ z_0 &= -q_0^4(i^3+k_0YT)B_{t,0},\\ (A_1)_{13} &= 32q_0^2k_0X^2Y^4\{4q_0^2Y\gamma+q_0^3(q_0-F)\}. \end{aligned}$$

Therefore $\deg P_{11}\le 213$. Its coefficient at

$$j^{48}i^{24}\eta^{37}X^{30}Y^{67}F^4\alpha^2\gamma$$

is exactly $2^{13}(3\beta)^{48}>0$. The minimum q_0 degree and the maximum displayed i degree select this coefficient uniquely in (4.6) times $(A_1)_{13}$; it is independent of K_0, M_C, M_F. Thus the exact degree on the actual numeral tuple is 213.

The standard-library checker passes normally and with Python assertions disabled. It checks the complete prefix, the octic conjugate identity, the weighted leading form, the nonzero degree certificate, coefficient hashes, deterministic rebuild, all eleven live inputs, and all 112 live arithmetic operations. Constructed Pell examples supplement the unbounded proof above; they do not test consistency or replace the inherited semantic calibration.

5 Seventeen integers: compressing congruence witnesses

The following constructions use unrestricted integer coordinates. They give the variable–degree pairs $(23,10)$, $(17,88)$, and $(16,184)$, with a $(16,192)$ sum-of-squares control. The first two improve the incoming $(24,10)$ and $(19,88)$ points [4, 3]; the last trades degree for another coordinate. Their literal circuits use respectively 131, 114, and 135 binary operations; the degree-192 control uses 134. Scalar arithmetic and squaring are counted. These are different tradeoffs from the much higher-degree ten-integer reductions. In particular, the 114-operation circuit is larger than the incoming 109-operation nineteen-integer circuit. No global minimum or historical priority is claimed.

All four use the same six fixed integers

$$\beta = 2^{1064921}-1, \qquad K_0, M_C, M_F, \qquad A_c = 2^{1113}, \quad B_c = 2^{666104}.$$

The first four are the entire source-34 `mixed-parity-unrounded-1064921` tuple; M_F already includes its shift by β. The last two are source71's determinant scales [9]. The actual six integers, compressed minimal little-endian bytes, finite native recipe, and complete raw hashes are supplied in the integer component's `coefficients/` directory. The raw encodings total 615 883 bytes. These fixed leaves are not additional unknowns. They satisfy

$$\beta\ge 17, \quad \beta\equiv 1 \pmod 3, \quad 0<M_C<\beta<M_F<2\beta, \quad K_0\ge 0, \quad A_c, B_c\ge 2.$$

5.1 Two inherited guards

Put $\mathrm{Pos}(p_1,p_2,p_3)=1+p_1^2+p_2^2+p_3(p_3+1)$. It takes exactly the positive integer values, by the three-square theorem applied to $4n-3$. The nearest-square joint-positivity lemma used in the incoming constructions is

$$x,y>0 \iff \exists t,p_1,p_2,p_3\in\mathbb{Z}: \quad xy-(t^2-x-y+1)^2=\mathrm{Pos}(p_1,p_2,p_3). \tag{5.1}$$

Indeed a positive left side forces $xy > 0$ and excludes both inputs being negative. Conversely, for $x, y > 0$, choose the nearer neighboring square to $x + y - 1$. Its squared error is less than $x + y - 1 \le xy$, leaving a positive integer represented by Pos.

For $\alpha = q - F - C$ define

$$d = (A_cF)(B_cC) - (q - A_cF - B_cC)^2.$$

The inequality $d > 0$ forces F, C, α, q to have one common nonzero sign. To see this, put $U = A_cF$, $B = B_cC$. Then U, B have a common sign and $|q - U - B| < \sqrt{UB} \le (|U| + |B|)/2$. Hence q has their sign and $|q| > (|U| + |B|)/2 \ge |F| + |C|$, which also fixes the sign of α. Source71's separate canonical-witness estimates prove completeness for this determinant; that calibrated existence restriction is inherited, not re-proved by finite sign tests.

5.2 The complete polynomial

The seventeen independent integer coordinates are

$$(J, C, z, f, h, r, v, w, \tau, \nu, \gamma, T, c, t, p_1, p_2, p_3).$$

The following are polynomial abbreviations only:

$$\begin{aligned}
&J_0 = 3J + 1, && q = \beta J_0 + 1, && W = 4(3w + 1),\\
&X = qW, && Y = (2v + 1)q^3, && E = XY,\\
&a = (X + 1)Y, && H = 4a + 3, && \Delta = a^2 + H,\\
&F = (K_0 + W)C - (q - 1)z, && \alpha = q - F - C.
\end{aligned}$$

Continue with

$$\begin{aligned}
&R = (q - F)(q^3 - q) - C(q^2 - 1) + (M_C + qM_F)J_0,\\
&k = R + 1 + hE, \qquad \eta = c - Yk, \qquad \zeta = k - \eta,\\
&I = (2r + 1)c^2, \qquad V = c(Tf - 1) - Rf^2,\\
&m = XY^2, \qquad B_0 = mk, \qquad s_0 = X + \gamma H, \qquad Z = \eta\zeta.
\end{aligned}$$

Define five residuals:

$$\begin{aligned}
A_1 &= \tau^2 + (2\tau - k)B_0 - 1,\\
A_2 &= 2acs_0 + s_0^2 - Hc^2 - 1,\\
A_3 &= I^2 - \Delta(f^2 - 1),\\
A_4 &= (V + \nu)^2 - I^2\nu(2V + \nu) - 1,\\
A_5 &= dZ - (t^2 - d - Z + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3).
\end{aligned} \tag{5.2}$$

The ordinary integer polynomial is

$$\boxed{P_{17} = A_1^2 + A_2^2 + A_3^2 + A_4^2 + A_5^2.} \tag{5.3}$$

The centered roots $\tau_{\mathrm{old}} = \tau + B_0$ and $y_{\mathrm{old}} = \nu + V$ give the identities

$$A_1 = \tau_{\mathrm{old}}^2 - m(m + 1)k^2 - 1, \qquad A_4 = I^2(V^2 - y_{\mathrm{old}}^2) + y_{\mathrm{old}}^2 - 1.$$

Also $D = ac + s_0$ gives $A_2 = D^2 - \Delta c^2 - 1$.

5.3 One quotient replaces two

Lemma 5.1 (Integer quotient compression)**.** *Suppose $c \neq 0$ and $c \mid f^2 - 1$. The simultaneous integer identities*

$$V = of - c = jc - R$$

are equivalent, after projection of j, o, to $V = c(Tf - 1) - Rf^2$ for one integer T.

Proof. From the old identities, $c \mid of + R$. The identity

$$f(of + R) - o(f^2 - 1) = o + Rf$$

shows that $T = (o+Rf)/c$ is integral and has the desired value of V. Conversely put $b = (f^2-1)/c$ and define

$$o = cT - Rf, \qquad j = Tf - 1 - Rb.$$

These are integers and restore both identities. The witness-restoration division introduces neither division nor a b coordinate into P_{17}. □

The order in which $c \mid f^2 - 1$ is established matters. Invoking an auxiliary-index theorem that already used the deleted congruences would be circular. Here the needed bootstrap is source70's signed orientation argument *before* its auxiliary-congruence step [8]. We spell out its dependency boundary.

At a P_{17} zero, $A_5 = 0$ gives $d > 0$ and $\eta\zeta > 0$. Write $Q = |q|$, $U = |X|$, $Z_0 = |Y|$, $K = |k|$, and $M = |E|$. The determinant, authentic mask bounds, and affine residue restrictions give

$$Q \geq 16, \quad Q(Q^2 - 1) < R < Q^4 - 2, \quad U \geq 4Q, \quad Z_0 \geq Q^3, \quad M > 3R + 1.$$

The positive product gives $K \geq 2$, $0 < |\eta| < K$, and $K(Z_0 - 1) < |c| < K(Z_0 + 1)$. Let $B = |a + 2|$ and $P = |2XY^2 + 1|$. They satisfy $P > B \geq 2$ and $B > R$. Classifying the first and main Pell norms gives

$$K = 2\psi_P(n_0), \qquad |c| = \psi_B(p), \qquad p > n_0.$$

The recurrence modulo M and $k = R + 1 + hE$ give $\pm 2n_0 \equiv R + 1 \pmod{M}$. The positive branch gives $n_0 \geq (R + 1)/2$, while the negative branch gives $n_0 > R$. Thus $p > n_0 \geq 24$; Pell growth yields

$$|c| > B(B^2 - 1)^2, \qquad |c| > 2p, \qquad |c| > 2R.$$

The inherited ordinary strong-rank lemma can now be applied to A_3. It gives $|f| = \chi_B(l)$ with $p \mid l$, without using either missing congruence. Therefore $\psi_B(p) \mid \psi_B(l)$ and

$$c \mid (B^2 - 1)\psi_B(l)^2 = f^2 - 1.$$

The precise retained source proof is `upstream/ORIENTATION_PROOF.md`, Sections 3–4.2; its auxiliary-congruence argument starts in Section 4.3.

Lemma 5.1 restores j, o. All other deleted coordinates are monic polynomial definitions, the two root shifts are integral, and the old positive guard triples can be restored by Pos. This gives a zero of the incoming calibrated 37-integer affine quartic [2]. Conversely, any zero of that source supplies the same divisibility, hence T; its transport row gives $F = (K_0 + W)C - (q - 1)z$ after the invertible change $\alpha = q - F - C$. Forget the monic coordinates, perform the root shifts, and use (5.1). Thus P_{17} has an integer zero exactly when that source does. This is a new existence equivalence on its full integer solution set, with no additional canonical restriction.

The exact verifier restores all 22 arithmetic rows of the literal parent: four match the new arithmetic residuals, one is $\pm R(cb - f^2 + 1)$, and all others vanish identically. It thereby exposes the sole use of the proved divisibility rather than silently treating b as an unknown.

6 Sixteen integers: merging an ordinary and a rational Pell root

Lemma 6.1 (Separated rational norm)**.** *Let $A, Q, S, V \in \mathbb{Z}$, with $A > 0$, $Q > 1$, and $S^2 > A$. Put $\delta = Q - 1$ and $B_{\text{num}} = S^2(QV^2 - 1)$. Then*

$$\exists n \in \mathbb{Z} : \quad [\delta(n^2 - A) - B_{\text{num}}]^2 - 4\delta A B_{\text{num}} = 0$$

if and only if there are integers u, y with $u^2 = A$ and $QV^2 - (Q-1)y^2 = 1$.

Proof. If $V = 0$, the displayed expression is a square plus the strictly positive integer $4\delta A S^2$. Thus any zero has $|V| \geq 1$. The rational number $B = B_{\text{num}}/\delta$ then satisfies $B \geq S^2 > A$. Dividing by δ^2 gives $(n^2 - A - B)^2 = 4AB$. Here $n \neq 0$, since $n = 0$ would give $A = B$. The identity

$$(n^2 + A - B)^2 - 4n^2 A = (n^2 - A - B)^2 - 4AB$$

shows that $u = (n^2 + A - B)/(2n)$ is a rational square root of the integer A, hence an integer. The rational $s = n - u$ satisfies $s^2 = B$, so $y = s/S$ is rational and obeys $QV^2 - (Q-1)y^2 = 1$.

Source69's adjacent-coefficient Pell descent makes this y integral. For completeness, if $y = a/d$ is reduced with $d > 0$, the norm gives $d^2 \mid Q - 1$. Make V, y positive by changing signs. Unless $V = 1$, the integral coordinate

$$V' = (2Q - 1)V - 2(Q - 1)y$$

and rational $y' = (2Q - 1)y - 2QV$ preserve the norm. The reduced denominator of y' remains d, since $2Q - 1 \equiv 1 \pmod{d}$. The inequalities $y > V$ and $QV > (Q - 1)y$ imply $-V < V' < V$; the norm excludes $V' = 0$. Strong induction on $|V|$ reaches $V = 1$, where $y = 1$ and $d = 1$. This is the inherited descent proof, applied to the new separated norm. Conversely, take $n = u + Sy$ and factor the norm. □

Delete τ, ν from the seventeen coordinates and add one unrestricted integer n. Keep every other abbreviation above and put

$$\begin{aligned} A_0 &= 1 + m(m+1)k^2, \qquad S = k(2m+1), \qquad \delta = I^2 - 1, \\ L &= \delta\big[n^2 + 2n(B_0 + SV) + 2B_0 SV - mk^2 - 1\big] + S^2(1 - V^2), \\ N &= L^2 - 4\delta A_0 S^2(I^2V^2 - 1). \end{aligned}$$

The new equation is the four-square polynomial

$$\boxed{P_{16} = A_2^2 + A_3^2 + A_5^2 + N^2.} \tag{6.1}$$

It is exactly the norm of Lemma 6.1 with $Q = I^2$ and $n_{\text{total}} = n + B_0 + SV$, because

$$\delta(n_{\text{total}}^2 - A_0) - S^2(I^2V^2 - 1) = L.$$

This centering cancels the two leading V^2 terms and lowers the degree.

The separation does not assume either missing norm. From $A_5 = 0$ alone, $d > 0$ and $\eta\zeta > 0$. The determinant and fixed coefficient bounds imply $|q| \geq 16$; the definitions give $|Y| \geq |q|^3$ and $|X| \geq 4|q|$. As η, ζ have the same sign, $|k| \geq 2$ and

$$|c| > |k|(|Y| - 1) > 1.$$

Hence $I^2 > 1$, while $|m| > 1$ gives $A_0 > 0$. Finally the exact identity

$$S^2 - 4A_0 = k^2 - 4 \geq 0 \tag{6.2}$$

proves $S^2 > A_0$. All hypotheses of the rational-norm lemma are therefore established before either deleted Pell root is extracted.

At a zero of P_{16}, the lemma supplies integral τ_{old} and y_{old}; put $\tau = \tau_{\mathrm{old}} - B_0$ and $\nu = y_{\mathrm{old}} - V$. This restores a zero of P_{17}. Conversely a P_{17} zero gives a P_{16} zero by $n = \tau + S\nu$. This proof covers arbitrary signed coordinates, and introduces no new divisibility restriction on canonical witnesses.

6.1 Exact degree certificates

For P_{17} the degrees of X, Y, E, a, R, k are $2, 4, 6, 6, 5, 7$; η, ζ have degree 11 and d has degree 4. The five residual bounds are $24, 14, 14, 14, 44$. Only the joint guard reaches degree 44. Its top part is $-\eta_{\mathrm{top}}^4$, with

$$\eta_{\mathrm{top}} = -2^4 3^8 \beta^7 h J^7 w v^2.$$

Consequently

$$[h^8 J^{56} w^8 v^{16}] P_{17} = 2^{32} 3^{64} \beta^{56} > 0,$$

proving exact degree 88 on the actual fixed coefficient slice.

For P_{16}, $\deg S = 17$ and $\deg V = 7$. The top part of L is $-S_{\mathrm{top}}^2 V_{\mathrm{top}}^2$, of degree 48; the second term of N has degree at most 94. Thus N has degree 96. Specifically,

$$S_{\mathrm{top}} = 2^8 3^{13} \beta^{11} h J^{11} w^2 v^3,$$
$$V_{\mathrm{top}} = f^2 (324 \beta^3 J^3 w C - 81 \beta^4 J^4 z),$$

and

$$[h^8 J^{120} w^{16} v^{24} z^8 f^{16}] P_{16} = 2^{64} 3^{136} \beta^{120} > 0.$$

The other squared rows have degrees at most $28, 28, 88$, proving exact degree 192. The direct uncentered weighted norm would have upper degree 216. Exact homogeneous-component arithmetic, rather than a naive gate bound, verifies both coefficients in the supplied auditor.

6.2 A degree-184 finalizer

Let $G = A_2^2 + A_3^2 + A_5^2$. Instead of $G + N^2$, define

$$\boxed{P_{16}^{\mathrm{unit}} = G + N + GN = (1 + G)(1 + N) - 1.} \tag{6.3}$$

For every integer assignment $G \geq 0$. If the new polynomial is zero, the positive integer $1 + G$ multiplies the integer $1 + N$ to give 1; therefore $G = N = 0$. Conversely these equalities make both polynomials zero. This exact zero-set equivalence needs no sign or congruence assumption about N.

The degree is $88 + 96 = 184$, certified by

$$[h^{12} J^{116} w^{16} v^{28} z^4 f^8] P_{16}^{\mathrm{unit}} = 2^{64} 3^{132} \beta^{116} > 0.$$

The expanded form $G + N + GN$ uses 135 binary operations, one more than the degree-192 control. The archive retains both explicit polynomials.

7 A lower-degree companion in twenty-three integers

This companion combines the incoming residue orientation with the paired Pell-root translations [2, 4]. Its independent coordinates are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, \nu, \gamma, o, j,\ E, a, I,\ t, p_1, p_2, p_3.$$

Here F, α, E, a, I are coordinates, not their expressions in the seventeen-variable equation. Define

$$\begin{aligned}
J_0 &= 3J + 1, & q &= \beta J_0 + 1, & W &= 4(3w + 1),\\
X &= qW, & Y &= a - E, & k &= \eta + \zeta,\\
c &= Yk + \eta, & V &= fo - c, & C &= q - F - \alpha,\\
H &= 4a + 3, & \Delta &= a^2 + H, & B_0 &= E(c - \eta),\\
s_0 &= X + \gamma H.
\end{aligned}$$

Use

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J_0$$

and the earlier definitions of d, $Z = \eta\zeta$, and A_5. The eleven residuals, whose squared sum defines P_{23}, are

$$\begin{aligned}
&a - E - (2v + 1)q^3, && E - X(a - E),\\
&\tau^2 + (2\tau - k)B_0 - 1, && 2acs_0 + s_0^2 - Hc^2 - 1,\\
&k - 1 - hE - R, && (K_0 + W)C - (q-1)z - F,\\
&V + k - 1 - cj - hE, && I - (2r + 1)c^2,\\
&I^2 - \Delta(f^2 - 1), && (V + \nu)^2 - I^2\nu(2V + \nu) - 1,\\
&A_5.
\end{aligned}$$

The count is $37 - 6 + 4 - 12 = 23$: replace the source affine quartic's six private guard coordinates by four, translate both roots, and delete twelve monic coordinates. Their exact restorations are

$$\begin{aligned}
N &= q^3, & X &= qW, & L &= q^2 - 1, & C_2 &= c^2,\\
F_2 &= f^2, & B_{\text{aux}} &= I^2, & g &= \nu(2V + \nu), & \Delta &= a^2 + 4a + 3,\\
D &= ac + s_0, & P &= k + B_0, & c &= (a - E)k + \eta, & V &= fo - c,
\end{aligned}$$

with $\tau_{\text{old}} = \tau + B_0$, $y_{\text{old}} = V + \nu$. Every arithmetic map is integral and polynomial. The auditor checks all 22 source arithmetic rows: twelve vanish identically and ten match the displayed arithmetic rows once each, up to sign. It also authenticates both old guard arguments.

The residual degrees are at most $4, 3, 4, 5, 4, 2, 3, 5, 4, 5, 4$. Only the I definition contains r, so

$$[r^2 a^4 \eta^4] P_{23} = 4.$$

Thus the degree is exactly ten, independently of the fixed ports.

Theorem 7.1 (Inherited calibration, new integer reductions)**.** *For each $P \in \{P_{23}, P_{17}, P_{16}, P_{16}^{\text{unit}}\}$, composition of the new reductions above with the named source native/compiler, Pell, and syntactic-refutation calibration theorems proves*

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{\dim P} : P(\mathbf{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}).$$

The theorem asserts the equivalence, not either side. The full native and logical calibration is inherited; the finite algebra/data checks are not an end-to-end Lean/Rocq or encoded internal-ZFC certificate. All four literal packets, inlined expressions, actual coefficients, exact degree receipts, source restorations, and shared DAGs are included. The verifier uses only Python's standard library and works with assertions disabled. This component's new steps have independent mathematical review within the AI collaboration; that is not external human peer review.

8 Reproducibility, audit coverage, and repository use

8.1 Complete literal artifacts

The JSON packets encode finite division-free arithmetic DAGs with a single output. The integer component additionally includes fully inlined residual expressions. The variable lists are part of the packet; temporary registers do not enlarge them. All large fixed coefficients are present as compressed minimal little-endian integer bytes, with their signs where needed and SHA-256 bindings. The nine-positive majorant is a supplied integer coefficient table rather than a runtime Laurent-series operation.

The principal files are:

Purpose	Path
Nine-positive literal polynomials	`nine/equations/`
Majorant and positive defect tables	`nine/data/`
Independent nine-positive proof and audit	`nine/certificates/`
Eleven-positive polynomial	`positive11/p11_positive_degree213.json`
Integer literal circuits	`integer/circuits/`
Integer inlined residual packets	`integer/equations/`
Expanded ordinary proofs	`positive11/PROOF.md`, `integer/PROOF.md`
Input provenance and formal status	`provenance/`, `PROOF_STATUS.md`
Release integrity and fresh transcripts	`MANIFEST.sha256`, `verification/`

8.2 Exact checks and diagnostics

The release driver `verify_all.py` invokes five audits in normal Python and with assertions disabled by `-O`. All active checks use explicit error conditions rather than relying on removable assertions. The audits have the following roles.

- The primary nine-positive audit verifies the literal regenerated circuits, source extraction, actual fixed coefficient recipes, topological order, liveness, operation counts, and cancellation-aware degrees. Independently written direct modular formula evaluations are transcription diagnostics.
- The independent H1868 audit imports no producer. It reconstructs all 22 source identities and 20,466 convolution interactions, the source height and binary population threshold, the 3035-term radicand, the 333-term corrected majorant, the 36,309-term positive shifted defect, and the unique highest-degree contribution.
- The local saturation audit checks 5472 arithmetic cases and constructs three complete Pell fixtures. These are diagnostic controls for the separate unbounded proof.
- The eleven-positive audit binds the exact circuit to the literal formulas, checks fixed coefficient bytes, expands the 1110-term double-centered norm in abstract variables, and verifies a nonzero degree-213 coefficient. Constructed Pell fixtures test the growth mechanism without standing in for its proof.
- The integer audit checks all source arithmetic restorations, the sole quotient-divisibility remainder, exact local norm identities, rational-root diagnostic cases, degree-attaining coefficients, fixed ports, and the complete literal DAGs. It does not assume degree attainment from an upper-bound propagation.

The positive majorant certificate is stronger than a collection of sampled inequalities: its exact translated polynomial identity and positive coefficients prove the bound throughout the stated

domain. Conversely, the modular evaluations and small Pell examples are not represented as proofs of the compiler semantics or of ZFC consistency.

From the root of an extracted archive, the full finite replay is

```
python3 verify_all.py --both
```

Only Python 3.10 or later and its standard library are needed. No symbolic algebra package is required for this replay. The generator and each verifier can also be run separately; the root README gives the exact commands. Compilation of this report needs only a conventional TeX Live installation:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

8.3 Integration and limitations

The archive is suitable for placement as another research contribution under the repository's Diophantine-consistency collection. It need not replace any currently certified equation or change an existing theorem statement. A reviewer can inspect the ordinary lemmas, replay the finite checks, and select an equation according to the domain and cost measure that matters for the next stage.

The independently implemented audits and reciprocal mathematical reviews were performed within the AI collaboration. They are not external human peer review. The finite data are authenticated against the supplied snapshot; the complete older native/compiler and logical-calibration chain remains an identified dependency. The number of variables, exact degrees, circuit counts, and coefficient payloads are concrete artifacts. Their global optimality is not established.

9 Alternative encodings and what remains open

The request permits constructions unrelated to K215 or a Jones universal polynomial. We examined that possibility as a separate design question. The repository already has a direct annotated checker for sealed ZFC proofs, explicit primitive-recursive programs for its syntax operations, and a complete numeric refutation checker [1]. Merely suggesting direct proof encoding would therefore repeat existing work.

A direct checker could provide a valuable route to the exact internal ZFC target without importing the numerical machine-calibration bridge. However, a checker program, an existence theorem from MRDP, and a fully instantiated small ordinary polynomial are different deliverables. To compete with the present equations, that route still needs a measured shared arithmetic graph, a complete explicit coefficient binding, and unbounded source/target correctness at the chosen witness domain. An unspecified program-index constant does not resolve those obligations.

The active repository reports a complete direct-checker program tree of 83,281 nodes, or 76,313 after its certified optimization, while its smaller shared quadratic graph remains unfinished [1]. Those are program-tree sizes, not counts of Diophantine variables. We have not converted them into an advertised smaller equation. Likewise, the ten-integer results of Sun are relevant comparison architecture [12], but do not by themselves materialize a smaller fixed consistency polynomial for this particular calculus.

The useful general lesson from the new constructions is local and concrete: when an arithmetic certificate already forces a square or a divisibility, its polynomial presentation can sometimes exploit that consequence without preserving the original witness parameterization. The soundness proof must establish the consequence before using it; completeness can rebuild suitable witnesses existentially. Positive Pell saturation, quotient recovery, and the guard-only root separation each

apply that principle at a different stage. They improve completed equations while leaving the compiler-independent route as an explicit open construction problem.

References

[1] Vladimir Reshetnikov and ProveIt contributors. *ProveIt: DiophantineConsistency research and incoming contributions*. Reviewed commit `c1a102936aca437e9999530a0111e6ddab56dc6d`, 24 October 2026. Pinned research directory. The source status, size plan, K215 status, and object-ZFC obligation are retained under `provenance/`.

[2] ProveIt incoming contribution. *Structural Reductions, Round 11*. `ConZFC_Structural_Reductions_Round11_2026-10-24.zip`, at the pinned repository's `docs/incoming/`. H1868 packing, norm saturation for the ten-integer construction, and the 37-integer affine quartic source.

[3] ProveIt incoming contribution. *ZFC Diophantine Joint Guards*. `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`. Joint positivity, integer low-degree alternatives, and the eleven-positive centered interfaces.

[4] ProveIt incoming contribution. *Budget and Centering*. `con-zfc-budget-and-centering-2026-10-23.zip`. Paired root translations and integer degree-ten comparison.

[5] ProveIt incoming contribution. *Radical Guard Descent / Cross-Guard Descent*. `con-zfc-radical-guard-descent-2026-10-23.zip`. Whole-radical majorants and coefficient/data recipes.

[6] ProveIt incoming contribution. *Standalone report*, `report.zip`, containing `report.tex`, at the pinned `docs/incoming/`. Its nine-positive degree-1 158 240, 811-operation entry is a reported comparison; this archive contains no executable companion.

[7] ProveIt contributors. *54-calibrated-carry* and the later compact54-sharp H2082 candidate. The complete coherent compact tuple, source adapter, inherited proof notes, and 440-operation parent DAG are retained under `nine/baseline/compact2082/`.

[8] ProveIt contributors. *70-residue-orientation*. Signed strong-rank recovery and the eleven-positive native Pell source. Relevant proof files are retained under `integer/upstream/` and `positive11/provenance/`.

[9] ProveIt contributors. *71-determinant-guards*. Positive nine-coordinate and calibrated determinant interfaces. Relevant proofs, scales, and notices are retained in the corresponding components. Source labels follow the pinned repository placements.

[10] ProveIt contributors. *ProductPell.lean*, the positive Pell certificate in the formalized Jones development. Pinned source file.

[11] James P. Jones. Universal Diophantine equation. *The Journal of Symbolic Logic* **47**(3), 549–571, 1982.

[12] Zhi-Wei Sun. *Ten unknowns for Hilbert's tenth problem over the integers*. arXiv:2609.01594, version 2, 2 September 2026. https://arxiv.org/abs/2609.01594.

[13] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher. *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*. arXiv:2506.20909, version 2, 2025. https://arxiv.org/abs/2506.20909.